



Data-Driven Functional Analysis of a Mammalian Neuron Type Connectome



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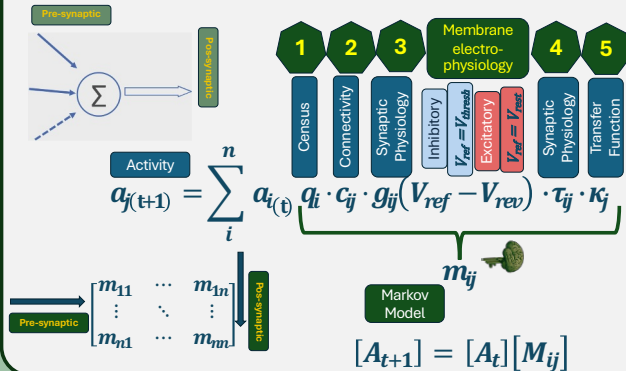
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Problem

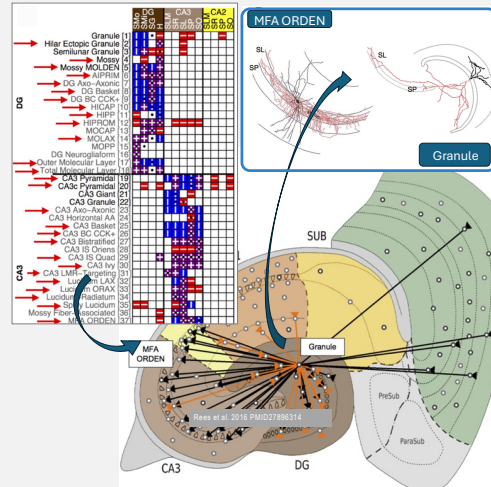
The enormous **complexity** of and **computational cost** for simulating biological neural networks is a major challenge in computational neuroscience. Translating this complexity into data-driven simulations requires availability of suitable experimental data and modeling feasibility

Objective

This work provides a computational framework founded on experimentally derived parameters to model hippocampal circuits at the crucial level of neuron types, enabling efficient simulations by reducing connectivity complexity without sacrificing biological accuracy



Hippocampus Neuron Type Classification



Hippocampome.org maps all known neuron types in the rodent hippocampal formation. V1.0 (2015) identified 122 neuron types with axons and dendrites distributed across 26 anatomical parcels

The resultant circuit consists of 38 excitatory and 84 inhibitory **neuron types** (nodes) interlinked by 3236 directional edges.

This comprehensive connectivity profile of the hippocampus and entorhinal cortex fosters insights on the functional operations at the computationally crucial level of **neuron types**

1 Census q_i

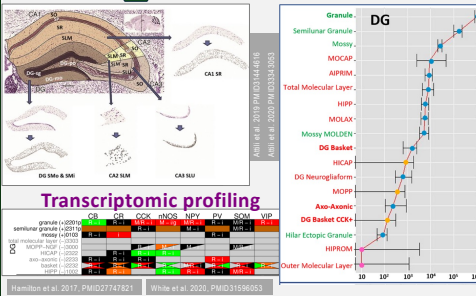
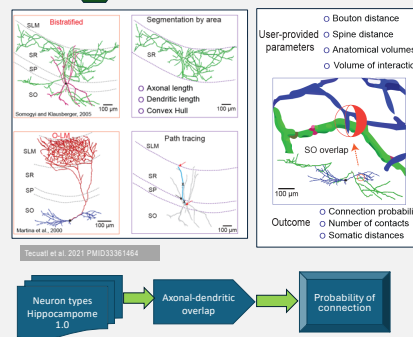


Image acquisition, correction, & analysis of neuron distributions

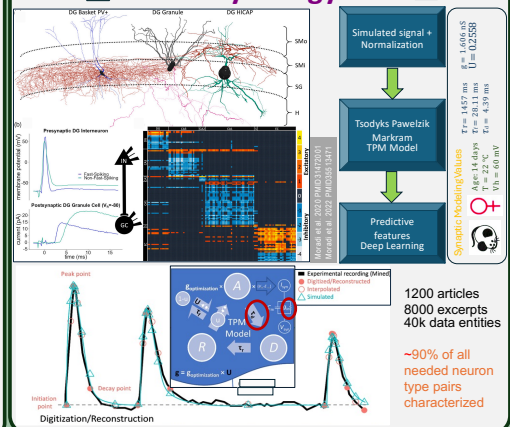


2 Connectivity C_{ij}



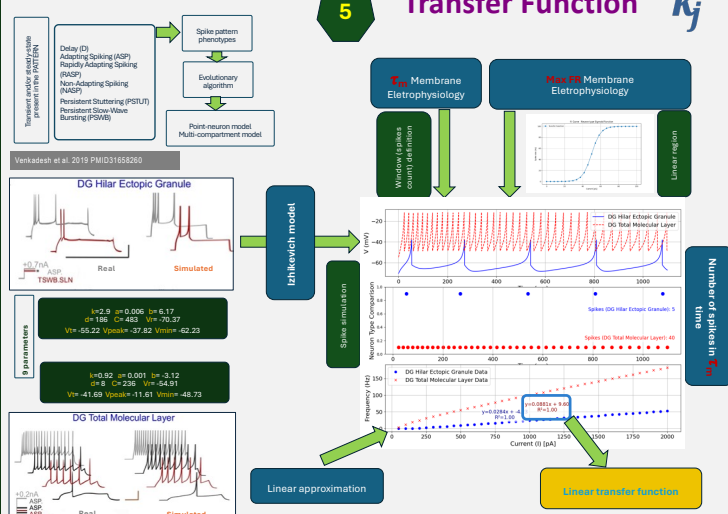
Connectivity stoichiometry of all directional neuron pairs. Computational geometry also derives the distances along the axonal path (affecting signal failure and delay) the dendritic path (affecting signal attenuation and spread)

3 g_{ij} Synaptic Physiology τ_{ij}



1200 articles
8000 excerpts
40k data entries
~90% of all needed neuron type pairs characterized

Izhikevich Model



Hippocampome.org utilizes a dynamic system framework to quantify the neuron type-level input-output function. Modeling the neuron type behavior captures the essence of its computational role within the spiking neural network while reducing the overall complexity to a simpler transfer function. This can be further simplified into a linear Markov model for graph-theoretic network analyses or substituted with a more realistic compartmental Hodgkin-Huxley representation for multi-scale modeling

Applications

